

Corporate ESG Rating Prediction Based on XGBoost-SHAP Interpretable Machine Learning Model

Abstract: The prediction of corporate ESG ratings is of paramount importance in augmenting the scientific rigor and precision of ESG investment decisions and steering corporate management of ESG-related risks. While machine learning methodologies have been extensively utilized in forecasting corporate behavior, their deployment in corporate ESG ratings remains relatively nascent and is often criticized for a lack of interpretability. This study develops a predictive model for corporate ESG ratings using an XGBoost algorithm enhanced with SHAP interpretability. The methodological framework incorporates SMOTE-ENN for handling class imbalance and a comprehensive optimization approach utilizing 3-fold cross-validation and randomized hyperparameter search. The model incorporates a comprehensive set of 15 indicators spanning four critical dimensions—financial performance, environmental impact, social responsibility, and corporate governance, using a dataset of Chinese A-share listed companies from 2013 to 2022. The model's predictive efficacy is subsequently elucidated, revealing that the XGBoost-SHAP framework achieves an accuracy and precision rate of 91.0% and 90.7%, respectively, with an F1-score and AUC value of 90.1% and 0.977, outperforming comparative models. The analysis underscores the significant influence of financial and non-financial factors on ESG rating predictions, with financial attributes exerting a relatively more pronounced impact than individual non-financial metrics. Tailored to the diverse objectives of ESG investors, this research further delineates a level definition model and a risk identification model, achieving predictive accuracies of 92.4% and 97.7%, respectively. The insights from this study furnish ESG investors with a robust foundation for enhancing investment outcomes and offer strategic guidance for corporations aiming to elevate their ESG performance.

Keywords: machine learning; XGBoost algorithm; SHAP; ESG rating prediction

1. Introduction

With the rapid global proliferation of ESG (Environmental, Social, and Governance), ESG ratings have assumed an increasingly pivotal role in augmenting corporate ESG performance and informing investors' strategic decision-making processes. Nevertheless, there are still problems in its use (Aaron K et al., 2016). From a corporate perspective, China has yet to establish a comprehensive ESG evaluation framework (Shen Hongtao et al., 2023), compounded by the opacity surrounding the specific assessment criteria employed by rating agencies. Identifying predictive indicators for ESG performance could provide valuable insights, guiding corporations to enhance their ESG profiles. For investors, the utility of ESG ratings is constrained by their time lag. Consequently, the accurate forecasting of future ESG ratings would empower investors to proactively formulate investment strategies, thereby enhancing risk mitigation capabilities and optimizing returns. Within this context, exploring predictive indicators for corporate ESG performance and establishing an effective ESG rating prediction model emerge as critical imperatives demanding practical resolution.

With the advancement of financial technology, many scholars have turned to machine learning techniques to predict corporate ESG ratings, aiming to enhance the scientific rigor and accuracy of ESG investment decision-making and to guide companies in managing ESG risks more effectively. D'Amato et al. (2021) established an ESG prediction model utilizing the random forest algorithm, demonstrating its robust predictive performance. Chowdhury F A M et al. (2023) employed six machine learning algorithms to construct an ESG rating prediction model, identifying the Random Forest algorithm as the top performer with an accuracy rate of

78.5%. David K L et al. (2024) leveraged Bloomberg's comprehensive dataset to establish a tailored ESG rating prediction model for Chinese companies, employing advanced machine learning techniques to enhance its predictive capabilities. Yan L et al. (2024) introduced an RCT model to address the issue of inconsistent information abundance in ESG-related features within company data, verifying its effectiveness and superiority in predicting ESG indicators. Wang Z et al. (2024) proposed a stacked generalization model using various machine learning algorithms, including Random Forest, as base learners and built an ESG rating prediction model based on this approach. Aggarwal D et al. (2024) utilized a hybrid model combining complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) and long short-term memory (LSTM) for ESG index prediction.

Nevertheless, machine learning algorithms commonly employed in predictive model construction, such as Random Forest, exhibit inherent limitations in processing large-scale datasets, thereby constraining the enhancement of predictive performance. In contrast, the Extreme Gradient Boosting Tree (XGBoost) algorithm, introduced by Chen et al. (2016), demonstrates superior capabilities in handling voluminous data. Empirical studies have substantiated its efficacy in addressing diverse financial problems, including default risk early warning systems(Lai Wenlong, 2023; Gong Yicheng et al.,2019; Xia Dong et al.,2023; Wang Jun et al.,2022) , credit risk assessment(Stefano Zedda, 2024; Rongrong Chen et al., 2020; Wang Kui et al., 2022) , financial fraud detection(Yunchuan Sun et al., 2024; Liu Yi, 2023; Ali Amal Al et al., 2023; Dalal Surjeet et al., 2022; Shimin LEI et al., 2020), and stock index futures forecasting(Wang Jujie et al., 2023).

Although machine learning algorithms offer the potential for enhanced predictive accuracy, they often suffer from limited interpretability, as understanding the influence of input features on output results remains a significant challenge. In contrast, the Shapley Additive exPlanation (SHAP) algorithm addresses this limitation by quantifying feature importance by computing each feature's absolute mean SHAP value. Furthermore, it enables the analysis of feature impacts on predictions by visualizing the relationship between feature values and their corresponding SHAP values. Owing to this distinctive capability, the SHAP algorithm has been widely adopted across various domains, however, its application in ESG rating prediction remains relatively underexplored.

In this study, we employ the SHAP algorithm to systematically evaluate and rank the importance of 31 influencing factors across four critical dimensions: financial performance, environmental impact, social responsibility, and corporate governance. Based on this analysis, we identify 15 key indicators that significantly influence ESG ratings. We utilize them as input features to construct an XGBoost-SHAP interpretable machine learning model for ESG rating prediction and subsequent interpretive analysis of the results. The proposed model offers dual benefits for both investors and corporate entities. For investors, it provides an intuitive understanding of each company's ESG performance and effectiveness through the generated ratings, thereby enhancing the scientific rigor and accuracy of ESG investment decision-making. The model serves as a strategic tool for ESG risk management for corporations by elucidating how specific input features influence the output results, enabling targeted improvements in ESG practices. Furthermore, to address the diverse objectives of investors utilizing ESG ratings, we establish two supplementary models: a level definition model and a risk identification model. These additional models enable investors to precisely determine a company's relative positioning within the ESG landscape and identify potential risks of suboptimal ESG performance. This comprehensive analytical framework allows investors to proactively adjust their investment strategies, thereby facilitating the optimization of their investment portfolios and maximizing returns.

The research contributions of this study are delineated as follows: First, this study constructs an interpretable ESG rating prediction model for companies based on the XGBoost-SHAP framework. While existing literature predominantly employs the Random Forest algorithm for ESG rating prediction, the XGBoost algorithm in this study demonstrates superior algorithmic efficiency, enhanced model performance, and excellent capability in handling large-scale datasets, thereby significantly improving predictive accuracy. Furthermore, the SHAP algorithm facilitates an interpretative analysis of the prediction results, offering valuable insights into how input features influence the output, thereby enhancing the transparency and explainability of the model. Second, this study advances the methodology for indicator selection. Unlike prior studies that predominantly rely on either financial indicators or environmental, social responsibility, and corporate governance (ESG) indicators in isolation, this research integrates both dimensions to construct a more comprehensive set of predictors. This approach enables a more robust exploration of the factors influencing ESG ratings, thereby improving the model's explanatory power. Third, this study enhances the applicability of ESG rating predictions by constructing multiple prediction models tailored to specific use cases. Recognizing the diverse objectives of investors in utilizing ESG ratings, this study establishes three models: a rank classification model, a level definition model, and a risk identification model. This multi-faceted approach ensures that the predictive outputs are more aligned with the varied needs of stakeholders, thereby increasing the practical utility of the research findings.

2. Basic methods and principles

2.1. XGBoost algorithm

XGBoost is an advanced ensemble learning algorithm designed to enhance gradient-boosted trees' computational efficiency and predictive performance. The algorithm operates by iteratively constructing decision trees to fit the residual errors of preceding models, thereby progressively refining the model's predictive accuracy. In the context of ESG rating prediction, the residual errors are typically derived by computing a specified loss function between a company's predicted ESG rating and its actual ESG rating, with the negative gradient of this loss function serving as the optimization target for subsequent tree construction.

The objective function of the XGBoost algorithm consists of two parts: the loss function and the regularization term. The loss function is shown in Equation (1), and the regularization term is shown in Equation (2).

$$l(y_i, \hat{y}_i^{(t)}) \approx l(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \quad (1)$$

$$\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T \omega_j^2 \quad (2)$$

In Eq. (1), g_i and h_i are the first and second-order derivatives of the loss function concerning $\hat{y}_i^{(t-1)}$, respectively, $f_t(x_i)$ is the predicted value of the t th tree for the i th sample. In Eq. (2), T is the number of leaf nodes, ω_j is the weight of the j th leaf node, γ and λ is the coefficient of the regularization term.

To solve the optimal solution efficiently, the XGBoost algorithm groups the samples belonging to the same leaf node into one group. It defines a new variable $q(x)$ to denote the leaf node index to which sample x is mapped. As a result, the objective function of the ESG rating

prediction model can be written as a quadratic function concerning the weights of the leaf nodes, as shown in Equation (3).

$$Obj^{(t)} \approx \sum_{i=1}^n \left[g_i \omega_{q(x_i)} + \frac{1}{2} h_i \omega_{q(x_i)}^2 \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T \omega_j^2 \quad (3)$$

Then, for the j th leaf node, its optimal weight and the corresponding objective function value are shown in Equation (4) and Equation (5), respectively.

$$\omega_j^* = - \frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \quad (4)$$

$$Obj^{(t)}(q) = - \frac{1}{2} \sum_{j=1}^T \frac{(\sum_{i \in I_j} g_i)^2}{\sum_{i \in I_j} h_i + \lambda} + \gamma T \quad (5)$$

where I_j is the set of samples contained in the j th leaf node.

Finally, this study uses a greedy algorithm. The tree is constructed by traversing all possible splits and choosing the one that makes the $Obj^{(t)}(q)$ smallest split.

2.2. SHAP algorithm

The SHAP algorithm leverages Shapley values to interpret machine learning models, aiming to allocate a contribution value to each feature for every sample. This approach facilitates local and global comprehension of the model's predictive behavior. The significance of a feature is quantified by computing the absolute mean of its corresponding SHAP values. Through the graphical representation of feature values against their SHAP values, one can discern the influence of each feature on the model's predictions. In addition, in a multi-categorization problem like predicting ESG rating, SHAP calculates a separate SHAP value for each level.

Assuming that there are N samples and M features, the SHAP value is defined as shown in Equation (6) for the j th feature of the i th sample.

$$\Phi_{ij} = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f_i(S \cup \{j\}) - f_i(S)] \quad (6)$$

Where F is the set of all features, S is the subset that does not contain feature i , $f_i(S)$ is the model's predicted value under the subset of features S , and is the marginal contribution of feature i .

Calculate the absolute mean of SHAP values of feature j in all samples as shown in Equation (7).

$$SHAP_j = \frac{1}{N} \sum_{i=1}^N |\Phi_{i,j}| \quad (7)$$

2.3. SMOTE-ENN mixed sampling method

The SMOTE-ENN hybrid sampling methodology, which integrates Synthetic Minority Oversampling Technique (SMOTE) and Edited Nearest Neighbours (ENN), is primarily employed to address class imbalance issues and enhance predictive accuracy. Within this sample equalization framework, the minority classes are initially augmented through SMOTE, thereby increasing the representation of minority class samples. Subsequently, the oversampled dataset undergoes a refinement process utilizing ENN to eliminate noise and ambiguous boundary samples, optimizing the classifier's performance.

2.4. Model evaluation indicators

(1) Accuracy

Accuracy is the proportion of all correctly predicted samples to the total sample. The calculation process in the rank classification model is shown in Equation (8).

$$Accuracy = \frac{\sum_{i=1}^5 C_{ii}}{\sum_{i=1}^5 \sum_{j=1}^5 C_{ij}} \quad (8)$$

(2) Precision · Recall and F1 Score

For each classification level i , the precision rate is the ratio of correctly predicted samples classified at level i to the total number of samples predicted at that level. The recall rate represents the proportion of accurately predicted samples among all samples actually belonging to level i . The F1 score, calculated as the harmonic mean of precision and recall, is a comprehensive metric for evaluating model performance at each classification level. Higher values of these metrics indicate superior model performance at the respective classification level. In the rank classification model, the calculation process is shown in Equations (9), (10), and (11).

$$Precision_i = \frac{C_{ii}}{\sum_{j=1}^5 C_{ji}} \quad (9)$$

$$Recall_i = \frac{C_{ii}}{\sum_{j=1}^5 C_{ij}} \quad (10)$$

$$F1Score_i = 2 \times \frac{Precision_i \times Recall_i}{Precision_i + Recall_i} \quad (11)$$

Recognizing that ESG rating prediction constitutes a multi-class classification problem and the class imbalance across rating levels in the test set samples, this study employs a weighted average approach to compute the aggregate precision, recall, and F1-score metrics, thereby facilitating a comprehensive evaluation of the model's overall performance. In the rank classification model, the calculation process is shown in Equation (12).

$$Precision / Recall / F1Score = \frac{\sum_{i=1}^5 (Precision_i / Recall_i / F1Score_i \times N_i)}{N} \quad (12)$$

Where $\text{Precision}_i/\text{Recall}_i/\text{F1Score}_i$ is the precision, recall, and F1 value for level i , N_i is the number of samples that are actually level i , and N is the total number of samples.

(3) AUC value and ROC curve

The Area Under the Curve (AUC) and Receiver Operating Characteristic (ROC) curves are critical metrics for assessing model performance. The ROC curve graphically represents the trade-off between the true positive rate and the false positive rate, thereby illustrating the predictive efficacy of the model. The AUC quantifies the area under the ROC curve and provides a scalar measure of the model's predictive capability. In multiclass classification, the One-vs-Rest (OvR) approach is predominantly employed. This method designates each class as positive while aggregating the remaining classes as negative. Individual ROC curves are subsequently generated for each class, and the corresponding AUC values are computed and ultimately averaged to yield a consolidated performance metric.

3. Modeling for ESG rating prediction

3.1. Establishment of data sets

This study utilizes the Huazheng ESG rating data, applying the following screening criteria: (1) the observation period spans from 2014 to 2023; (2) all ST, ST*, and PT companies are excluded; (3) companies with missing or anomalous data are removed. Following these criteria, a final sample of 819 companies, comprising 8,190 company-year observations, is obtained. The Huazheng ESG rating agency employs a nine-tier grading system: AAA, AA, A, BBB, BB, B, CCC, CC, and C. Notably, companies rated BBB, BB, and B constitute a substantial proportion of the sample, reflecting the distributional characteristics of the financial market, where most companies exhibit moderate ESG performance, while exceptionally high or low ESG performers are relatively scarce. To address the imbalanced distribution of ratings, this study consolidates the top-tier ratings (AAA, AA, and A) into a single 'A' category and the lowest-tier ratings (CCC, CC, and C) into a 'C' category, resulting in a five-tier classification system: A, BBB, BB, B, and C. Furthermore, to mitigate class imbalance and data noise issues, this study employs the SMOTE-ENN hybrid sampling method for dataset balancing (Guoli Mo et al., 2024). The percentage of frequencies after sample equalization are shown in **Table 1**.

Table 1

Percentage of frequencies after sample equalization.

Rank	Frequency	Percentage (%)
A	2975	44.087
BBB	1695	25.119
BB	374	5.542
B	319	4.727
C	1385	20.525
Total	6748	100.0

3.2. Feature Engineering

Feature Engineering is a key step in machine learning that aims to improve a model's

predictive performance by filtering out the most useful features from the original features.

3.2.1. Original features

Given that rating agencies primarily evaluate the environmental and social impacts of a company's production and business activities, along with the associated financial risks and returns (Delgado Ceballos Javier et al., 2023), and considering that empirical studies have demonstrated the significance of specific financial metrics, such as EBIT, in influencing corporate ESG ratings, this study adopts a comprehensive approach by selecting ESG rating determinants across both financial and non-financial dimensions (environmental, social responsibility, and corporate governance) for the period 2013–2022. Drawing on the ESG rating prediction framework established in prior research (D'Amato et al., 2021), four financial indicators—PE, ROE, LEV, and EBIT—were selected as proxies for the financial dimension, with data sourced from the CSMAR database. In alignment with the environmental performance metrics established in existing studies, nine indicators, including the presence of an environmental protection concept, environmental goals, and the adoption of an environmental management system, were chosen to represent the environmental dimension, with data also obtained from CSMAR. Based on the disclosure practices in corporate social responsibility (CSR) reports, 12 social responsibility indicators were selected, such as adherence to the Global Reporting Initiative (GRI) standards, disclosure of shareholder rights protection, and disclosure of creditor rights protection. These data were extracted from the Basic Information Sheet of Listed Companies' Social Responsibility Reports. Furthermore, referencing the corporate governance performance metrics constructed in prior research, six indicators, including InsInvestorProp, IndDirectorRatio, and ConcurrentPosition, were selected to represent the corporate governance dimension, with data sourced from CSMAR. **Table 2** provides an introduction to the definition of the indicators.

Table 2

Definition of the indicators.

Dimensions	Indicators	definition
Finance	PE	Ratio of stock price to earnings per share
	ROE	Ratio of net profit to average net assets
	LEV	Total liabilities to total assets
	EBIT	Sum of net income, interest expense and income tax
Environment	EPtConcept	Whether the company has an environmental concept
	EPGoal	Whether the company has environmental protection objectives
	EPManSysSchema	Whether the company has adopted an

		environmental protection management system
	EPeduTrain	Whether the company has conducted education and training on environmental protection
	EPSpecialAct	Whether the company has special behavior for environmental protection
	EnvEventEmergMech	Whether the company has adopted an emergency response mechanism for environmental incidents
	ThreeSimultaneity	Whether or not the company has a “three simultaneous” system in place
	EPHonorReward	Whether the company has received honors or awards for environmental protection
	ISO14001	Whether the company has passed the ISO14001 certification
	GRI	Whether the company discloses information with reference to GRI
	ShareholdersProtection	Whether the company discloses shareholder rights protection
Social Responsibility	CreditorProtection	Whether the company discloses creditor rights protection
	StaffProtection	Whether the company discloses staff rights protection
	DeliveryProtection	Whether the company discloses delivery rights protection
	CustomerProtection	Whether the company discloses customer and consumer rights protection
	EnvironmentProtection	Whether the company discloses environmental and sustainable development
	PublicRelations	Whether the company discloses public relations and social causes
	SystemConstruction	Whether the company discloses social

Corporate Governance		responsibility system construction and improvement measures
	WorkSafety	Whether the company discloses the content of production safety
	Deficiency	Whether the company discloses the deficiencies of the company
	IsMandatoryDisclose	Willingness of the company to disclose announcements
	InsInvestorProp	Ratio of number of shares held by institutional investors to total share capital
	IndDirectorRatio	Ratio of the number of independent directors to the total number of board members
	ConcurrentPosition	Whether the positions of chairman and general manager of the company are held by the same individual
	LargestHolderRate	Ratio of the number of shares held by the largest shareholder to total share capital
	Boardsize	Number of members of the board of directors of the company
	Mngmhldn	Ratio of number of shares held by executives to total share capital

The extant literature lacks a definitive consensus regarding which factors exert a more pronounced influence on ESG ratings. Consequently, it is imperative to initially incorporate a comprehensive array of potential determinants, followed by a rigorous screening process to identify those with the most substantial impact. Moreover, the aforementioned 31 indicators are not mutually independent, and this interdependency can be effectively mitigated by applying the SHAP method for feature selection.

3.2.2. Feature selection

In this study, the SHAP algorithm is employed for feature selection, which significantly reduces model complexity and computational resource consumption by filtering out feature variables that exhibit negligible contributions to prediction outcomes. This approach enhances the model's generalization capability and predictive accuracy and optimizes its overall performance. Initially, the original features are fed into the XGBoost model, and their importance is ranked based on the mean absolute SHAP values, as illustrated in **Figure 1**. The analysis reveals that EBIT exerts the most substantial influence on ESG ratings, with a mean absolute SHAP value of 0.7405, followed by the IsMandatoryDisclose factor, which has a mean absolute SHAP value of 0.3139. In contrast, 16 factors, including SystemConstruction and

WorkSafety, demonstrate mean absolute SHAP values below 0.1, indicating minimal impact on ESG ratings, which can be considered negligible. Consequently, this study selects the top 15 influential factors as input features for the XGBoost model. These factors encompass both financial and non-financial dimensions and include EBIT, IsMandatoryDisclose, LEV, PE, IndDirectorRatio, InsInvestorProp, Mngmhldn, LargestHolderRate, ROC, EPHonorReward, DeliveryProtection, Boardsize, ISO14001, EPManSysSchema, and CreditorProtection.

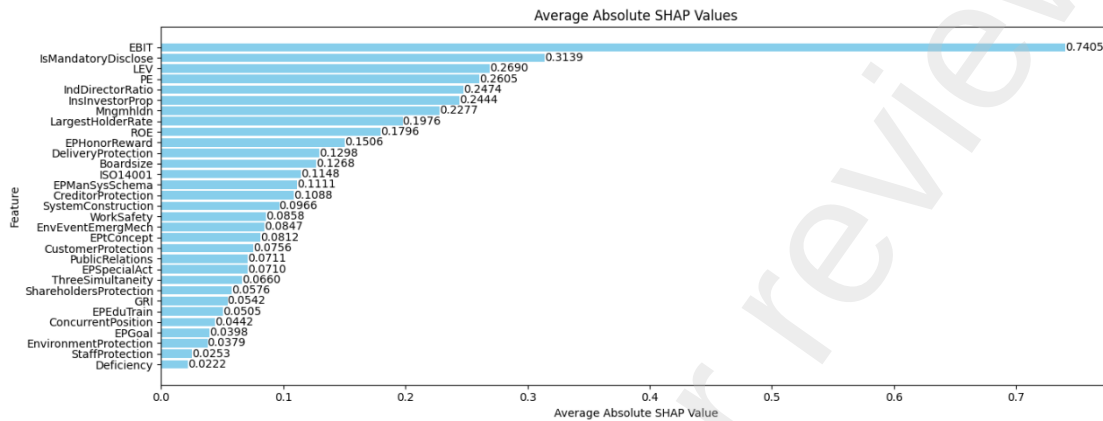


Figure 1 Importance ranking before feature selection

3.3.Optimization of parameters

The optimization of the XGBoost model's performance hinges on the adjustment of hyperparameters(Adil Mehdary et al., 2024), including max_depth, learning_rate, n_estimators, subsample, colsample_bytree, gamma, and min_child_weight. Specifically, max_depth governs the maximum depth of individual trees, while learning_rate dictates the shrinkage weighting applied to each tree's contribution. The n_estimators parameter determines the total number of trees in the ensemble, and the subsample regulates the fraction of observations utilized for training each tree. Concurrently, colsample_bytree controls the proportion of features sampled for constructing each tree. The gamma parameter establishes the minimum loss reduction required to execute a split, and min_child_weight specifies the minimum sum of instance weights necessary to form a new leaf node. In this study, a 3-fold cross-validation approach combined with a randomized search strategy was employed to identify the optimal hyperparameter configuration, as delineated in Table 3.

Table 3

Optimal hyperparameter configuration.

Hyperparameters	Value
max_depth	7
learning_rate	0.1
n_estimators	200
subsample	1.0
colsample_bytree	1.0
gamma	0
min_child_weight	3

4. Model prediction results and interpretive analysis

4.1. Analysis of model prediction results

This study partitioned the balanced dataset into training and testing subsets at an 8:2 ratio. The ESG rating prediction model is established utilizing the XGBoost algorithm and the training subset. For the purpose of rank classification modeling, ESG ratings are categorized into five distinct tiers: A, BBB, BB, B, and C. This classification framework enables investors to assess and compare the ESG-related performance across different companies effectively.

The test set samples are input into the trained model to generate the final prediction outcomes, as illustrated by the confusion matrix presented in **Figure 2**. In this matrix, the rows correspond to the actual grades, while the columns denote the predicted grades. Each element C_{ij} within the matrix quantifies the number of samples that are true of grade i but are classified as grade j . The numerical labels 1, 2, 3, 4, and 5 represent the grades C, B, BB, BBB, and A, respectively.

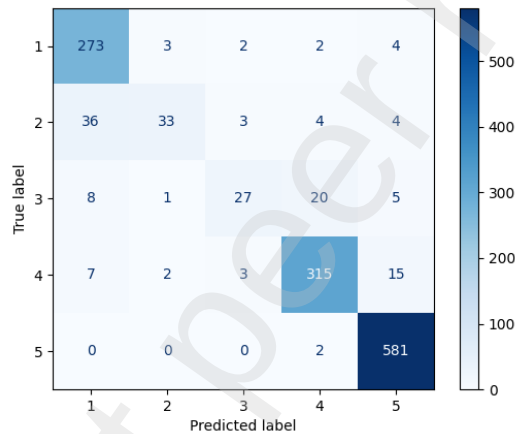


Figure 2 Confusion matrix of rank classification model

Based on the confusion matrix and Eqs. (8)-(11), the prediction performance evaluation metrics for each class were computed, as presented in **Table 4**. Out of the 1,350 samples in the test set, 1,229 were correctly predicted, yielding an overall accuracy of 91.0%. The detailed classification results indicate a non-negligible probability of misclassification across all grades. However, the misclassification probabilities for grades A, BBB, and C are relatively low, whereas those for BB and B are notably higher.

Table 4

Evaluation indicator values for each rank.

Rank	Accuracy	Precision	Recall	F1 Score
A	91.0%	95.4%	99.7%	97.5%
BBB		91.8%	92.1%	92.0%
BB		77.1%	44.3%	56.3%
B		84.6%	41.3%	55.5%
C		84.3%	96.1%	89.8%

By Equation (12), the comprehensive predictive performance metrics of the rank classification model are delineated in Table 5, accompanied by the ROC curve illustrated in Figure 3. The tabulated data reveal that the XGBoost rank classification model achieves an accuracy rate of 91.0%, a precision rate of 90.7%, a recall rate of 91.0%, and an F1 score of 90.1%. These metrics underscore the model's robust predictive capability and proficiency in accurately categorizing the five ESG ratings. Furthermore, the AUC value of 0.977, coupled with the ROC curve's proximity to the upper left, substantiates the model's exceptional classification efficacy.

Table 5

Evaluation indicator values for XGBoost rank classification model.

	Accuracy	Precision	Recall	F1 Score	AUC
XGBoost	91.0%	90.7%	91.0%	90.1%	0.977

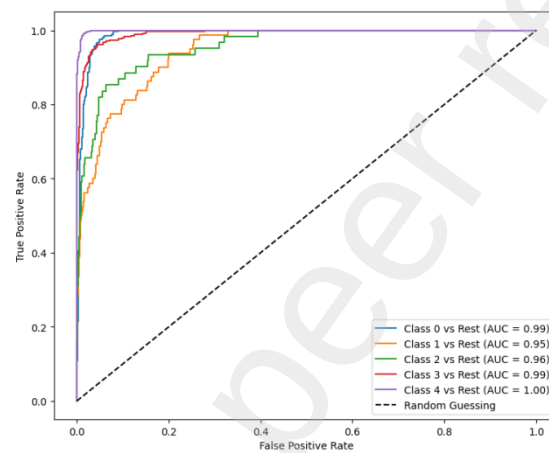


Figure 3 ROC curve of XGBoost rank classification model

4.2. Comparative analysis of model prediction effects

In order to comprehensively evaluate the model's predictive accuracy, this study utilizes two alternative machine learning approaches, namely the Gradient Boosting Decision Trees (GBDT) model and the CatBoost model, to forecast ESG ratings. The analysis is conducted on a balanced sample, adhering to the same procedural steps and evaluation metrics applied in assessing the XGBoost model's performance.

(1) Comparison model 1: GBDT

GBDT is an ensemble learning algorithm extensively employed in regression and classification tasks. As a boosting-based approach, it iteratively constructs multiple weak learners to approximate the optimal solution progressively. Ultimately, it aggregates the predictions of these weak learners to form a robust and accurate prediction from a strong learner.

The performance evaluation metrics of the GBDT model's predictive capabilities are presented in the first row of Table 6, with the corresponding ROC curve illustrated in Figure 4(a). The GBDT algorithm demonstrates predictive accuracy for ESG ratings, achieving an accuracy rate of 85.0%. Furthermore, the model exhibits an overall precision of 84.9%, a recall rate of 85.0%, an F1 score of 83.0%, and an AUC value of 0.951.

(2) Comparison model 2: CatBoost

CatBoost, an advanced machine learning algorithm built upon the GBDT, has undergone extensive optimization and refinement. It demonstrates exceptional proficiency in processing datasets with a high dimensionality of categorical features. Notably, the algorithm innovatively incorporates ordered target statistics, effectively addressing the issue of target leakage commonly encountered in conventional categorical feature processing methods. This methodological advancement enhances the model's predictive accuracy and robustness and maintains computational efficiency without compromising performance.

The predictive performance metrics of the CatBoost model are presented in the second row of **Table 6**, with the corresponding ROC curves illustrated in **Figure 4(b)**. The CatBoost algorithm achieved an accuracy of 87.7%, an overall precision of 87.5%, a recall of 87.7%, an F1 score of 86.2%, and an AUC value of 0.968.

The predictive efficacy of the XGBoost rank classification model surpasses that of the comparative models. Consequently, the XGBoost model in this study demonstrates superior performance in forecasting ESG ratings.

Table 6

Evaluation indicator values of comparison model.

Model	Accuracy	Precision	Recall	F1 Score	AUC
GBDT	85.0%	84.9%	85.0%	83.0%	0.951
CatBoost	87.7%	87.5%	87.7%	86.2%	0.968

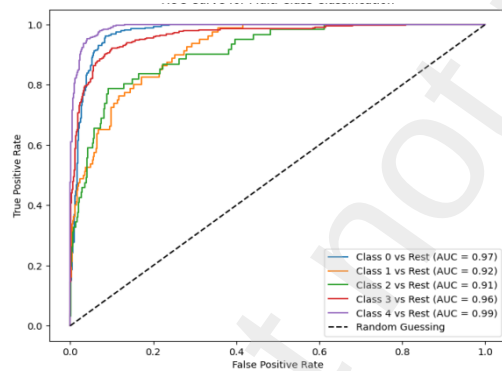


Figure 4(a) ROC curve for GBDT model

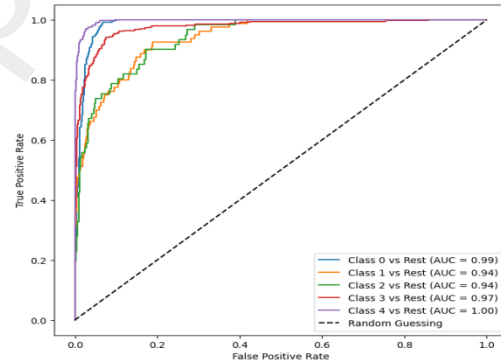


Figure 4(b) ROC curve for CatBoost model

4.3. Interpretive analysis of prediction results

4.3.1. Feature importance analysis

After establishing an ESG rating prediction model utilizing the XGBoost algorithm, this study incorporates the SHAP algorithm to elucidate the contribution of each feature to the predictive model. By applying SHAP, we derive the SHAP values for each model feature and subsequently rank their mean absolute values. This ranking elucidates the relative importance of each feature in the model's predictive capability, with higher-ranked features exerting a more substantial influence on the model's output. As illustrated in **Figure 5**, the mean absolute SHAP value for each feature exceeds 0.1, indicating that all features contribute to some extent to the prediction of ESG ratings. Specifically, financial features constitute approximately 37.0% of the

total feature importance, followed by environmental features at 15.4%, social responsibility features at 17.9%, and corporate governance features at 29.7%. This distribution underscores that financial and non-financial factors significantly impact ESG ratings, with financial features exerting a relatively more significant influence than individual non-financial dimensions. Consequently, financial factors are indispensable for companies aiming to enhance their ESG performance. Furthermore, EBIT, IsMandatoryDisclose, LEV, Mngmhldn, and EPHonorReward exhibit relatively higher importance, with mean absolute SHAP values exceeding 0.3. Features including InsInvestorProp, PE, IndDirectorRatio, ROE, LargestHolderRate, and DeliveryProtection are of moderate importance, while the remaining features contribute less significantly to the prediction of ESG ratings. This analysis highlights the critical role of financial metrics in ESG rating predictions and provides valuable insights for stakeholders aiming to improve corporate ESG performance through targeted strategic initiatives.

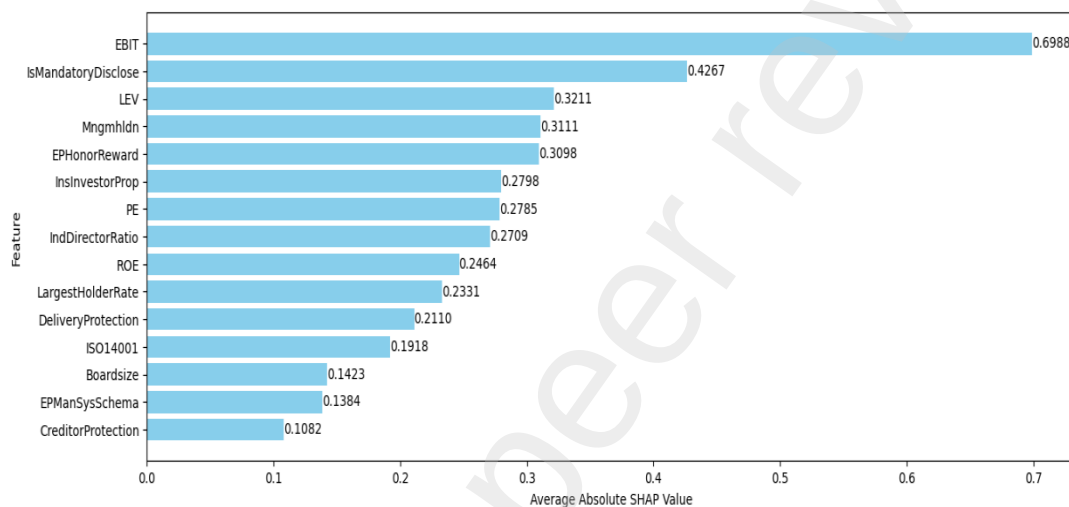


Figure 5 Ranking of feature importance

4.3.2. Feature dependency analysis

This study employs the SHAP algorithm to generate SHAP dependency plots, thereby facilitating an in-depth examination of how features with relatively high importance weights influence the predictive outcomes of ESG ratings.

Figure 6(a) illustrates that EBIT sample values positively correlate with SHAP values for grades A and BBB. In contrast, a negative correlation is observed for grades BB, B, and C. This suggests that companies with higher EBIT are more likely to be assigned higher ratings by rating agencies, while those with lower EBIT tend to receive lower ratings. In **Figure 6(b)**, the distribution of sample points indicates that companies with a lower propensity to disclose social responsibility announcements are more likely to be rated lower. In contrast, those with higher disclosure willingness tend to receive higher ratings. **Figure 6(c)** demonstrates that the SHAP dependency plot for LEV across all rating grades is non-linear. Specifically, for grades A and BBB, the relationship between LEV sample values and SHAP values follows an "inverted U" shape, while for grades BB, B, and C, it exhibits a "U" shape. This implies that rating agencies are more inclined to assign higher ratings to companies with moderate leverage ratios. In contrast, companies with excessively high or low leverage ratios are likelier to be rated lower. As depicted in the sample point distribution in **Figure 6(d)**, a higher executive shareholding ratio is associated with a greater likelihood of receiving lower ratings. In comparison, a lower ratio tends to correlate with higher ratings. Finally, **Figure 6(e)** reveals that companies receiving more

environmental protection honors or awards are more likely to be assigned higher ratings. In contrast, those with fewer such recognitions tend to receive lower ratings.

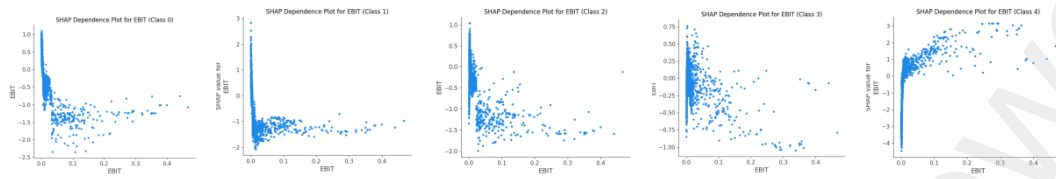


Figure 6(a) SHAP dependency charts for each level of EBIT

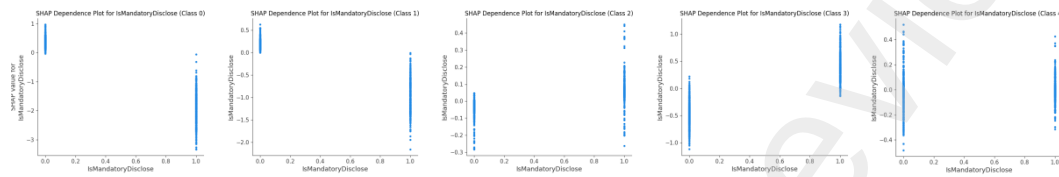


Figure 6(b) SHAP dependency charts for each level of IsMandatoryDisclose

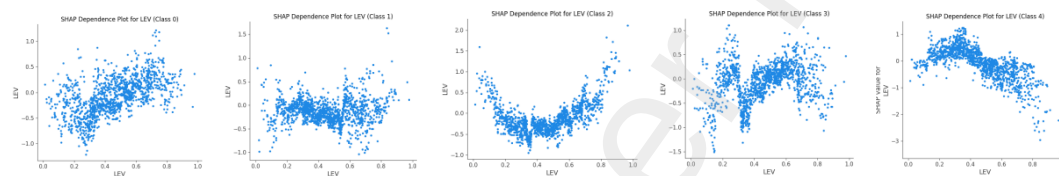


Figure 6(c) SHAP dependency charts for each level of LEV

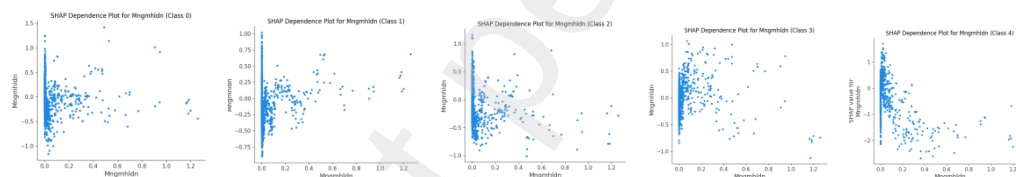


Figure 6(d) SHAP dependency charts for each level of Mngmhldn

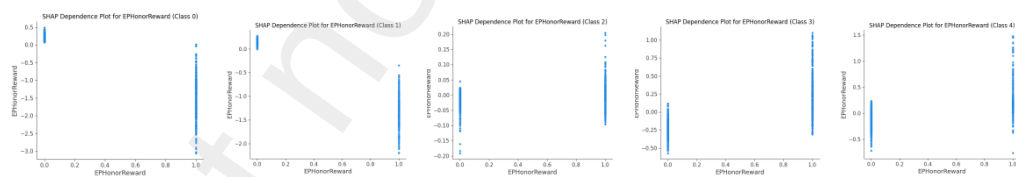


Figure 6(e) SHAP dependency charts for each level of EPonorReward

5. Further research

To enhance the applicability of ESG rating prediction outcomes, this research establishes two models—The level Definition Model and the Risk Identification Model—tailored to the divergent application objectives of investors, *ceteris paribus*. The Level Definition Model aids investors in ascertaining a company's relative standing within the ESG domain, whereas the Risk Identification Model facilitates the discernment of a company's ESG-related risks. In formulating the Level Definition Model, corporate ESG performance is stratified into three tiers: Leader, Average, and Laggard. Concurrently, the Risk Identification Model bifurcates corporate ESG performance into two classifications: High, indicative of commendable ESG performance, and Low, indicative of subpar ESG performance, thereby enabling investors to recalibrate their investment strategies to optimize returns preemptively. In constructing the Level Definition

Model, this study draws upon the classifications delineated on the official website of the CSI, amalgamating the four echelons of AAA, AA, A, and BBB into the Leader category, signifying the apex of ESG performance. The subsequent three echelons—BB, B, and CCC—are consolidated into the Average category, while the remaining two echelons—CC and C—are aggregated into the Laggard category, denoting the nadir of ESG performance. For the Risk Identification Model, the Leader and Average categories are collectively designated High, emblematic of robust ESG performance. In contrast, the Laggard category is designated as Low, emblematic of deficient ESG performance.

5.1. Level definition model

This study categorizes ESG performance levels into three classifications within the level definition model: Leader, Average, and Laggard. The aforementioned SMOTE-ENN hybrid sampling methodology is employed to address the inherent challenges of sample imbalance and noisy data. The resultant frequency distributions of the processed categories are presented in Table 7.

Table 7

Percentage of frequencies after sample equalization.

Level	Frequency	Percentage (%)
Leader	5956	36.515
Average	3861	23.671
Laggard	6494	39.814
Total	16311	100.0

Upon completing the sample processing, this study partitions the dataset into training and test sets at an 8:2 ratio. Subsequently, a classification model is constructed utilizing the XGBoost algorithm, which is trained on the training set. The test set is then fed into the trained model, and the predictive outcomes are delineated in the confusion matrix presented in Figure 7. Categories 1, 2, and 3 represent the laggard, average, and leader levels, respectively.

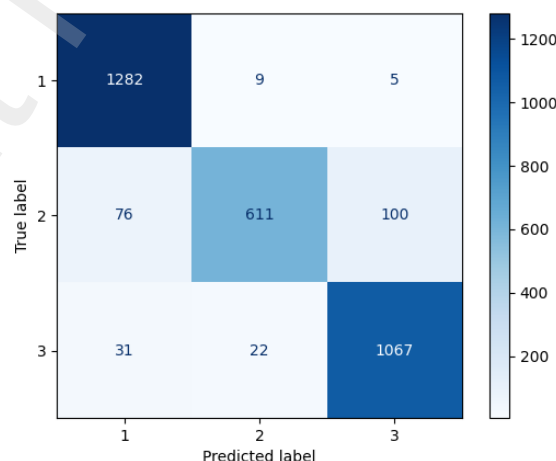


Figure 7 Confusion matrix of level definition model

By the methodological framework and computational formulae for assessing the predictive

efficacy of the XGBoost rank classification model, an analysis of the confusion matrix about the prediction outcomes of the aforementioned level definition model was conducted. This analysis yielded the following metrics indicative of the model's predictive performance: an accuracy rate of 92.4%, an aggregate precision of 92.6%, a recall rate of 92.4%, an F1 score of 92.2%, and an AUC value of 0.986. The ROC curves corresponding to the level definition model are illustrated in **Figure 8**.

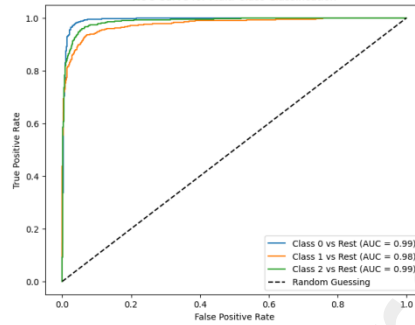


Figure 8 ROC curve of XGBoost level definition model

As illustrated in the figure, the ROC curve exhibits a pronounced proximity to the upper left corner, suggesting that the level definition model demonstrates high accuracy in providing investors with the company's relative positioning within the ESG domain.

5.2. Risk identification model

In the risk identification model established in this study, corporate ESG performance is classified into two distinct categories: ESG good and ESG poor. The study employs a combined SMOTE-ENN sampling technique to address the challenges of sample imbalance and noisy data. Due to space constraints, the frequency distribution table of the processed categories is not presented here. Following the sample processing phase, the dataset is partitioned into training and test sets in an 8:2 ratio. The risk identification model is subsequently constructed using the XGBoost algorithm, with the training set serving as the basis for model establishment. The test set is then utilized to evaluate the model's performance, with the prediction outcomes illustrated in the confusion matrix presented in **Figure 9**. In this matrix, categories 0 and 1 correspond to ESG underperformance and ESG good performance, respectively.

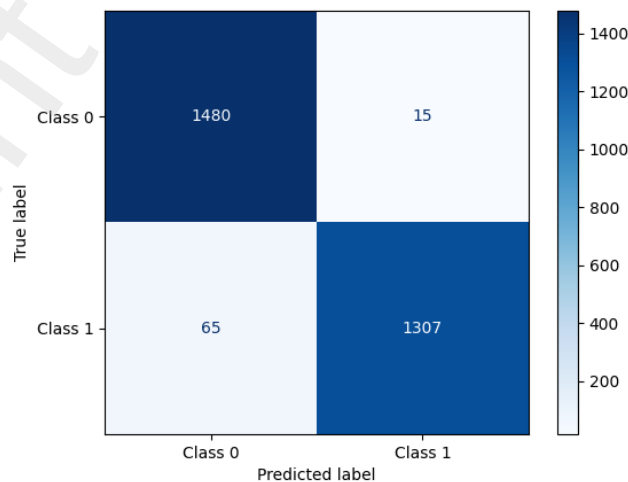


Figure 9 Confusion matrix of risk identification model

Through an analysis of the confusion matrix derived from the risk identification model's predictive outcomes, the final model's performance metrics were quantified as follows: accuracy, recall, and the F1 score each registered at 97.2%, while precision registered at 97.3%. The model's discriminative capability was further evidenced by an AUC value of 0.997. The ROC curve is illustrated in **Figure 10**.

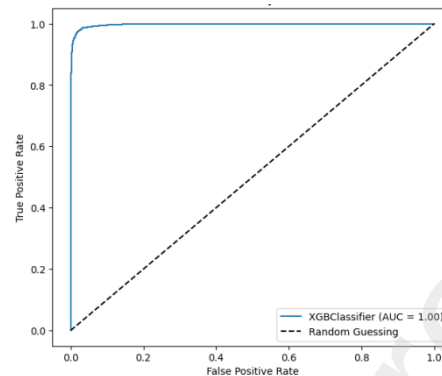


Figure 10 ROC curve of XGBoost risk identification model

The figure illustrates that the ROC curve is pronounced near the upper left, indicating a superior predictive capability. This observation underscores a significant enhancement in the risk identification model's accuracy relative to the rank classification and level definition models. Consequently, investors are afforded a more precise mechanism for ascertaining the presence of ESG risks within companies.

6. Conclusion and prospect

To enhance the scientific rigor and precision of ESG investment decision-making and to facilitate corporate management of ESG risks, this study introduces a predictive model for corporate ESG ratings utilizing the XGBoost algorithm and SHAP interpretability framework, yielding several key findings. First, in terms of model performance evaluation, the XGBoost-based rank classification model demonstrates superior predictive capabilities, achieving an accuracy and recall rate of 91.0%, overall precision of 90.7%, F1 score of 90.1%, and AUC value of 0.977, outperforming comparison models. This robust performance underscores the efficacy of the XGBoost algorithm in ESG rating prediction tasks. Second, the analysis reveals that financial and non-financial factors significantly influence ESG ratings, with financial indicators demonstrating greater predictive power than individual non-financial metrics. This finding highlights the critical importance of financial management in enhancing corporate ESG performance. Furthermore, the study investigates the relative importance of various predictive features, providing valuable insights for companies seeking to improve their ESG ratings. Third, to address diverse investor needs in utilizing ESG ratings, this research establishes three distinct model frameworks under consistent methodological conditions: a level definition model with 92.4% accuracy for assessing companies' relative ESG positioning and a risk identification model achieving 97.2% accuracy in detecting potential ESG performance risks.

In summary, this study proposes the following recommendations: (1) Investors should leverage machine learning methodologies to forecast forthcoming ESG ratings, thereby enabling the formulation of proactive investment strategies. Furthermore, during the strategy development process, investors should select appropriate predictive models tailored to their specific requirements, ensuring the applicability and reliability of model predictions. This

approach enhances risk management capabilities and optimizes investment returns. (2) For corporations, enhancing ESG performance necessitates a comprehensive approach that integrates financial, environmental, social responsibility, and corporate governance dimensions. Financial characteristics should be accorded due importance. Additionally, factors such as EBIT, disclosure willingness, gearing ratio, and executive shareholding ratio should be incorporated into future ESG rating assessments to provide a more holistic evaluation. (3) For rating agencies, there is a critical need to enhance the transparency of evaluation criteria in ESG performance assessments. This transparency will assist corporations in identifying specific areas for improvement in their ESG initiatives. Concurrently, regulatory bodies must intensify their oversight of corporate disclosures to ensure greater accuracy in ESG reporting.

Furthermore, companies operating within heavily polluting industries are inherently engaged in production and business activities detrimental to environmental sustainability and pose risks to life and health safety. Consequently, these companies exhibit suboptimal performance in ESG metrics, thereby attracting heightened scrutiny during the ESG investment evaluation process. In light of this, future research could enhance the granularity of analysis by segmenting the sample into two distinct cohorts: companies within heavily polluting industries and those outside such sectors. This bifurcation would facilitate the establishment of separate ESG rating prediction models tailored to the unique characteristics of each group. Particular emphasis should be placed on refining the model's predictive accuracy for heavily polluting industry companies through targeted improvements in indicator selection methodologies and algorithmic optimization techniques.

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Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.